Extended results to Approximate and exact reformulations of a capacity expansion model for single-allocation hub-and-spoke networks with congestion

S.1. Table S-1. Comparisons of related studies.

Reference	Modeling viewpoint									Methodological viewpoint				
	Network topology		Decision problem		Congestion		Model			Linear approximation		Exact method		Heuristic method
	HSN	Multi-commodity/other networks	Network design	Network expansion	Hub/ node	Hub link/arc	MILP	Convex MINLP	Non-convex MINLP	Error non-guarantee	Error guarantee	SOCP	Decomposition or others	
De Camargo et al. (2009)	✓		✓		✓			✓					✓	
Elhedhli and Wu (2010)	✓		✓		✓				✓	✓				✓
De Camargo et al. (2011)	✓		✓		✓			✓					✓	
Miranda et al. (2011)		✓		✓		✓		✓					✓	
Liu and Wang (2015)		✓		✓		✓			✓	✓			✓	
Sun and Schonfeld (2015)		✓		✓	✓			✓					✓	
Kian and Kargar (2016)	✓		✓		✓			✓				✓		
Paraskevopoulos et al. (2016)		✓		✓	✓				✓			✓		✓
Bärmann et al. (2017)		✓		✓			✓						✓	
Azizi et al. (2018)	✓		✓		✓				✓	✓			✓	✓
Hu et al. (2018)	✓		✓		✓	✓			✓					✓
Alkaabneh et al. (2019)	✓		✓		✓			✓		✓				✓
Mohammadi et al. (2019)	✓		✓		✓	✓			✓	✓				✓
Najy and Diabat (2020)	✓		✓		✓		✓						✓	
Karimi-Mamaghan et al. (2020)	✓	✓	✓		✓	✓			✓	✓				✓
Bhatt et al. (2021)	✓		✓		✓				✓			✓		
Şafak et al. (2022)		✓		✓					✓			✓	✓	
Zhang and Liu (2022)	✓			✓			✓						✓	
This work	✓			✓	✓	✓			✓		✓	✓		

S.2. Proof of Proposition 2

Proof. Given any $S_k \in \left[2^{b-1}, 2^b\right]$, $b = 1, 2, ..., |B_k| - 2$, $\forall k \in H$, the following equation system (S-1) can be constructed based on the H-LA reformulation.

$$\begin{cases} \sum_{l \in L} r_{k,l} = \frac{2^{b-1}}{1+2^{b-1}} \cdot g_{k,b-1}^0 + \frac{2^b}{1+2^b} \cdot g_{k,b}^0, \forall k \in H \\ g_{k,b-1}^0 + g_{k,b}^0 = 1, \forall k \in H \end{cases} \tag{S-1}$$

By solving equation system (S-1), we obtain that $g_{k,b-1}^0 = \frac{\left[2^b - \left(1+2^b\right) \cdot \sum_{l \in L} r_{k,l}\right] \cdot \left(1+2^{b-1}\right)}{2^b - 2^{b-1}}$ and $g_{k,b}^0 = \frac{\left[\left(1+2^{b-1}\right) \cdot \sum_{l \in L} r_{k,l} - 2^{b-1}\right] \cdot \left(1+2^b\right)}{2^b - 2^{b-1}}$. Then, $\hat{S}_k$ can be correspondingly expressed as $\hat{S}_k = \left(1+2^{b-1}+2^b+2^{b-1} \cdot 2^b\right) \cdot \sum_{l \in L} r_{k,l} - 2^{b-1} \cdot 2^b, \forall k \in H$.

Subsequently, it is proved that $\varepsilon_k^1 = \left|\frac{S_k - \hat{S}_k}{S_k}\right| = \frac{\hat{S}_k - S_k}{S_k}, \forall k \in H$. Denoting $\hat{S}_k - S_k = \frac{-\left[\left(1+2^{b-1}\right) \cdot \sum_{l \in L} r_{k,l} - 2^{b-1}\right] \cdot \left[\left(1+2^b\right) \cdot \sum_{l \in L} r_{k,l} - 2^b\right]}{1 - \sum_{l \in L} r_{k,l}}$ ($\forall k \in H$) and taking the derivative of $\hat{S}_k - S_k$ on $\sum_{l \in L} r_{k,l}$, we can derive that $\hat{S}_k - S_k$ is concave when $\sum_{l \in L} r_{k,l} \in \left[\frac{2^{b-1}}{1+2^{b-1}}, \frac{2^b}{1+2^b}\right]$. This means that it has a maximum value and a minimum value. It is noted that its minimum value is equal to 0 when $\sum_{l \in L} r_{k,l} = \frac{2^{b-1}}{1+2^{b-1}}$ or $\sum_{l \in L} r_{k,l} = \frac{2^b}{1+2^b}$. Thus, $\hat{S}_k - S_k \geq 0$. Again, since $S_k > 0$, $\varepsilon_k^1 = \left|\frac{S_k - \hat{S}_k}{S_k}\right| = \frac{\hat{S}_k - S_k}{S_k}, \forall k \in H$.

Then, we can further construct the following equation (S-2).

$$\frac{\hat{S}_k - S_k}{S_k} = -\left(1+2^{b-1}\right) \cdot \left(1+2^b\right) \cdot \sum_{l \in L} r_{k,l} + \left(2^{b-1} + 2^b + 2 \cdot 2^{b-1} \cdot 2^b\right) - \frac{2^{b-1} \cdot 2^b}{\sum_{l \in L} r_{k,l}}, \forall k \in H \tag{S-2}$$

Taking the first-order derivative of (S-2) on $\sum_{l \in L} r_{k,l}$, we obtain that

$$\frac{d\left(\frac{\hat{S}_k - S_k}{S_k}\right)}{d\left(\sum_{l \in L} r_{k,l}\right)} = -\left(1+2^{b-1}\right) \cdot \left(1+2^b\right) + \frac{2^{b-1} \cdot 2^b}{\left(\sum_{l \in L} r_{k,l}\right)^2}, \forall k \in H.$$

Let the first-order derivative of (S-2) be 0, we can derive that $\sum_{l \in L} r_{k,l} = \sqrt{\frac{2^{b-1} \cdot 2^b}{\left(1+2^{b-1}\right) \cdot \left(1+2^b\right)}}$, $\forall k \in H$. Taking the second-order derivative of (S-2) on $\sum_{l\in L} r_{k,l}$, we can demonstrate that

$$\frac{d^2\left(\frac{\hat{S}_k - S_k}{S_k}\right)}{d\left(\sum_{l\in L} r_{k,l}\right)} = -\frac{2\cdot 2^{b-1}\cdot 2^b}{\left(\sum_{l\in L} r_{k,l}\right)^3} < 0, \forall k\in H .$$

Therefore, $\frac{\hat{S}_k - S_k}{S_k}$ achieves the maximum value at $\sum_{l\in L} r_{k,l} = \sqrt{\frac{2^{b-1}\cdot 2^b}{\left(1+2^{b-1}\right)\cdot\left(1+2^b\right)}}$ ($\forall k\in H$), namely

$$\sup_{\sum_{l\in L} r_{k,l}\in\left[\frac{2^{b-1}}{1+2^{b-1}},\frac{2^b}{1+2^b}\right]}\left[\frac{\hat{S}_k - S_k}{S_k}\right] = \left(2^{b-1}+2^b+2\cdot 2^{b-1}\cdot 2^b\right) - 2\cdot\sqrt{2^{b-1}\cdot 2^b\cdot\left(1+2^{b-1}\right)\cdot\left(1+2^b\right)}, \forall k\in H .$$

Then, it is obtained that $\varepsilon_k^1 \le \left(2^{b-1}+2^b+2\cdot 2^{b-1}\cdot 2^b\right) - 2\cdot\sqrt{2^{b-1}\cdot 2^b\cdot\left(1+2^{b-1}\right)\cdot\left(1+2^b\right)}$ ($b\ge 1$). □

S.3. Proof of Proposition 3

Proof. Note that the congestion cost for hubs can be essentially represented as the term $\sum_{k\in H} c_k^0\cdot\hat{S}_k$ subjected to additional constraints in the H-LA reformulation. In the original model [M1], the congestion cost for hubs can be expressed as the term $\sum_{k\in H} c_k^0\cdot S_k$. Since $\hat{S}_k - S_k \ge 0$, which has been proved in Proposition 2, $\sum_{k\in H} c_k^0\cdot\hat{S}_k \ge \sum_{k\in H} c_k^0\cdot S_k$. Therefore, the solution to the H-LA reformulation provides the upper bound of the congestion cost for hubs, compared to the original model [M1]. □

S.4. Proof of Proposition 4

Proof. Since $\sum_{l\in L} z_k^l = 1$ and $z_k^l\in\{0,1\}$, it must guarantee that for each k, we obtain $z_k^l = 1$ for some $l\in L$ and $z_k^{l^*} = 0$ for all other $l^*\in L\backslash\{l\}$. Then, it is derived that $\sum_{l\in L}\Gamma_k^l\cdot z_k^l \in\left\{\Gamma_k^1,\Gamma_k^2,...,\Gamma_k^{|L|}\right\}$, $\forall k\in H$. Furthermore, it is easy to demonstrate that $1\big/\left(\sum_{l\in L}\Gamma_k^l\cdot z_k^l\right)\in\left\{1/\Gamma_k^1, 1/\Gamma_k^2,...,1/\Gamma_k^{|L|}\right\}$. Meanwhile, it is observed that $\sum_{l\in L}\left(1\big/\Gamma_k^l\right)\cdot z_k^l\in\left\{1/\Gamma_k^1, 1/\Gamma_k^2,...,1/\Gamma_k^{|L|}\right\}$ based on the fact that $\sum_{l\in L} z_k^l = 1$ and $z_k^l\in\{0,1\}$. These suggest that $1\big/\left(\sum_{l\in L}\Gamma_k^l\cdot z_k^l\right)$ and $\sum_{l\in L}\left(1\big/\Gamma_k^l\right)\cdot z_k^l$ take the value from the same set. Finally, it is shown that $1\big/\left(\sum_{l\in L}\Gamma_k^l\cdot z_k^l\right) = 1\big/\Gamma_k^{l^*}$ and $\sum_{l\in L}\left(1\big/\Gamma_k^l\right)\cdot z_k^l = 1\big/\Gamma_k^{l^*}$ when $z_k^{l^*} = 1$ ($l^*\in L$). Therefore, $1\big/\left(\sum_{l\in L}\Gamma_k^l\cdot z_k^l\right) = \sum_{l\in L}\left(1\big/\Gamma_k^l\right)\cdot z_k^l$, $\forall k\in H$. Then, we can derive that

$$\sum_{i\in N} O_i\cdot x_{i,k}\Big/\sum_{l\in L}\Gamma_k^l\cdot z_k^l = \left(\sum_{i\in N} O_i\cdot x_{i,k}\right)\cdot\left[1\Big/\left(\sum_{l\in L}\Gamma_k^l\cdot z_k^l\right)\right] = \left(\sum_{i\in N} O_i\cdot x_{i,k}\right)\cdot\left[\sum_{l\in L}\left(1\big/\Gamma_k^l\right)\cdot z_k^l\right]$$
$$= \sum_{i\in N}\sum_{l\in L} O_i\cdot\left(1\big/\Gamma_k^l\right)\cdot x_{i,k}\cdot z_k^l$$ □

S.5. Proof of Proposition 5

Proof. For the proof of this Proposition, we can refer to Theorem 1 in Fügenschuh et al. (2015). □

S.6. Proof of Proposition 6

Proof. For a given hub link $k-m$, the set $\zeta_{k,m}$ consists of the following points: $\left(\Lambda_{k,m}^{1}, \ln\left(\Lambda_{k,m}^{1}\right)\right)$, $\left(\Lambda_{k,m}^{2}, \ln\left(\Lambda_{k,m}^{2}\right)\right)$, …, $\left(\Lambda_{k,m}^{|Q|}, \ln\left(\Lambda_{k,m}^{|Q|}\right)\right)$. It is easy to show that these points are feasible for inequalities (69) and (70). These points can actually be expressed as the intersection of inequalities from (69) and (70).

However, each intersection of two arbitrary inequalities may lead to an infeasible point because such a point is cut off by some other inequalities from (69) and (70). To identify such points that are outside the convex hull, we check the following two cases.

Case 1. It represents the selection of two arbitrary inequalities from (69). For the convenience of expression, we define coefficients $\theta_{k,m}^{q} = \dfrac{\ln\left(\Lambda_{k,m}^{q}\right) - \ln\left(\Lambda_{k,m}^{q-1}\right)}{\Lambda_{k,m}^{q} - \Lambda_{k,m}^{q-1}}$ and $\vartheta_{k,m}^{q} = -\theta_{k,m}^{q} \cdot \Lambda_{k,m}^{q-1} + \ln\left(\Lambda_{k,m}^{q-1}\right)$ for $q = 2,...,|Q|$. Then, equation system (S-3) is given, where $q^*, q^{**} \in \{2,...,|Q|\}$ and $q^* \neq q^{**}$.

$$\begin{cases} lu_{k,m} = \theta_{k,m}^{q^*} \cdot \left(\sum\limits_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right) + \vartheta_{k,m}^{q^*} \\ lu_{k,m} = \theta_{k,m}^{q^{**}} \cdot \left(\sum\limits_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right) + \vartheta_{k,m}^{q^{**}} \end{cases} \tag{S-3}$$

(1) Assume $|q^* - q^{**}| = 1$, without loss of generality, let $q^{**} = q^* + 1$, $q^* \in \{2,...,|Q|-1\}$. Then, the solution of (S-3) is $\left(\Lambda_{k,m}^{q^*}, \ln\left(\Lambda_{k,m}^{q^*}\right)\right)$, $q^* \in \{2,...,|Q|-1\}$. Obviously, these solutions are points in $\zeta_{k,m}$.

(2) Assume $|q^* - q^{**}| > 1$, without loss of generality, let $q^{**} = q^* + qq$, $qq > 1$. Then, the solution of (S-3) is $\left(\dfrac{\vartheta_{k,m}^{q^{**}} - \vartheta_{k,m}^{q^*}}{\theta_{k,m}^{q^*} - \theta_{k,m}^{q^{**}}}, \dfrac{\theta_{k,m}^{q^*} \cdot \vartheta_{k,m}^{q^{**}} - \theta_{k,m}^{q^{**}} \cdot \vartheta_{k,m}^{q^*}}{\theta_{k,m}^{q^*} - \theta_{k,m}^{q^{**}}} \right)$. Obviously, it is easy to see that such a solution violates at least one inequality from (69): $lu_{k,m} \leq \theta_{k,m}^{q} \cdot \left(\sum\limits_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right) + \vartheta_{k,m}^{q}$, $q \in \{2,...,|Q|\} / \{q^*, q^{**}\}$. For example, let $qq = 2$, the obtained solution violates the inequality $lu_{k,m} \leq \theta_{k,m}^{q^*+1} \cdot \left(\sum\limits_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right) + \vartheta_{k,m}^{q^*+1}$. Thus, this solution does not belong to $\zeta_{k,m}$.

Case 2. It represents the selection of one arbitrary inequality from (69) and the one from (70). Then, equation system (S-4) is given, where $q \in \{2,...,|Q|\}$.

$$\begin{cases} lu_{k,m} = \theta_{k,m}^{q} \cdot \left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right) + \vartheta_{k,m}^{q} \\ lu_{k,m} = \dfrac{\ln\left(\Lambda_{k,m}^{|Q|}\right) - \ln\left(\Lambda_{k,m}^{1}\right)}{\Lambda_{k,m}^{|Q|} - \Lambda_{k,m}^{1}} \cdot \left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} - \Lambda_{k,m}^{1} \right) + \ln\left(\Lambda_{k,m}^{1}\right) \end{cases} \tag{S-4}$$

(1) When $q = 1$, the solution of (S-4) is $\left(\Lambda_{k,m}^{1}, \ln\left(\Lambda_{k,m}^{1}\right)\right)$, which is a point in $\zeta_{k,m}$.

(2) When $q = |Q|$, the solution of (S-4) is $\left(\Lambda_{k,m}^{|Q|}, \ln\left(\Lambda_{k,m}^{|Q|}\right)\right)$, which is a point in $\zeta_{k,m}$.

(3) When $q \in \{2, ..., |Q|-1\}$, the solution of (S-4) violates at least one inequality from (69): $lu_{k,m} \le \theta_{k,m}^{q^*} \cdot \left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right) + \vartheta_{k,m}^{q^*}$, $q^* \ne q$ and $q^* \in \{2, ..., |Q|\}$, which is similar to that of (2) in Case 1. Thus, the solution of (S-4) does not belong to $\zeta_{k,m}$. □

S.7. Proof of Proposition 7

Proof. Obvious from the construction. □

S.8. Proof of Proposition 10

Proof. First, based on Remark 4, it is obtained that $\left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q}\right)^{\beta} = \sum_{q \in Q} \left(\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q}$. Second, considering that $\sum_{q \in Q} u_{k,m}^{q} = 1$ and $u_{k,m}^{q} \in \{0,1\}$, $\sum_{q \in Q} \left(\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q} \in \left\{ \left(\Lambda_{k,m}^{1}\right)^{\beta}, \left(\Lambda_{k,m}^{2}\right)^{\beta}, ..., \left(\Lambda_{k,m}^{|Q|}\right)^{\beta} \right\}$, $\forall k, m \in H, k \ne m$. Furthermore, it is easy to demonstrate that $1 \Big/ \left[\sum_{q \in Q} \left(\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q} \right] \in \left\{ \left(1/\Lambda_{k,m}^{1}\right)^{\beta}, \left(1/\Lambda_{k,m}^{2}\right)^{\beta}, ..., \left(1/\Lambda_{k,m}^{|Q|}\right)^{\beta} \right\}$. Meanwhile, it is worth noting that $\sum_{q \in Q} \left(1/\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q} \in \left\{ \left(1/\Lambda_{k,m}^{1}\right)^{\beta}, \left(1/\Lambda_{k,m}^{2}\right)^{\beta}, ..., \left(1/\Lambda_{k,m}^{|Q|}\right)^{\beta} \right\}$, $\forall k, m \in H, k \ne m$. This result suggests that $1 \Big/ \left[\left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right)^{\beta} \right]$ and $\sum_{q \in Q} \left(1/\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q}$ have the same set that they take the corresponding value. Finally, it is proven that $1 \Big/ \left[\left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right)^{\beta} \right] = \sum_{q \in Q} \left(1/\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q}$, $\forall k, m \in H, k \ne m$. This is because that $\sum_{q \in Q} \left(1/\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q} = \left(1/\Lambda_{k,m}^{q^*}\right)^{\beta}$ and $1 \Big/ \left[\left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right)^{\beta} \right] = \left(1/\Lambda_{k,m}^{q^*}\right)^{\beta}$ when $u_{k,m}^{q^*} = 1$ ($q^* \in Q$). Then, we can give that

$$\begin{aligned} E_{k,m} \Big/ \left[\left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right)^{\beta} \right] &= E_{k,m} \cdot \left[1 \Big/ \left(\sum_{q \in Q} \Lambda_{k,m}^{q} \cdot u_{k,m}^{q} \right)^{\beta} \right] = E_{k,m} \cdot \left[\sum_{q \in Q} \left(1/\Lambda_{k,m}^{q}\right)^{\beta} \cdot u_{k,m}^{q} \right] \\ &= \sum_{q \in Q} \left(1/\Lambda_{k,m}^{q}\right)^{\beta} \cdot E_{k,m} \cdot u_{k,m}^{q}, \forall k, m \in H, k \ne m \end{aligned} \quad \square$$

S.9. Proof of Proposition 11

Proof. Note that the term $\sum_{k \in H} \sum_{\substack{m \in H \\ k \ne m}} c_{k,m}^{1} \cdot \Xi_{k,m} \cdot \left[\sum_{i \in N} y_{k,m}^{i} + \sum_{q \in Q} \phi \cdot \left(1 \Big/ \Lambda_{k,m}^{q} \right)^{\beta} \cdot \Omega_{k,m}^{q} \right]$ subjected to additional constraints in the HL-LA-2 reformulation represents the congestion cost for hub links, and it can be equivalently expressed as the term $\sum_{k\in H}\sum_{\substack{m\in H\\k\neq m}} c_{k,m}^{1}\cdot\Xi_{k,m}\cdot\left[\sum_{i\in N} y_{k,m}^{i}+\sum_{q\in Q}\phi\cdot\left(\frac{1}{\Lambda_{k,m}^{q}}\right)^{\beta}\cdot\hat{E}_{k,m}\cdot u_{k,m}^{q}\right]$. In the original model [M1], the congestion cost for hub links can be expressed as the term $\sum_{k\in H}\sum_{\substack{m\in H\\k\neq m}} c_{k,m}^{1}\cdot\Xi_{k,m}\cdot\left[\sum_{i\in N} y_{k,m}^{i}+\sum_{q\in Q}\phi\cdot\left(\frac{1}{\Lambda_{k,m}^{q}}\right)^{\beta}\cdot E_{k,m}\cdot u_{k,m}^{q}\right]$. Since $\hat{E}_{k,m}-E_{k,m}\geq 0$, which has been proved in Proposition 9, $\sum_{q\in Q}\phi\cdot\left(\frac{1}{\Lambda_{k,m}^{q}}\right)^{\beta}\cdot\hat{E}_{k,m}\cdot u_{k,m}^{q}\geq\sum_{q\in Q}\phi\cdot\left(\frac{1}{\Lambda_{k,m}^{q}}\right)^{\beta}\cdot E_{k,m}\cdot u_{k,m}^{q}$. Therefore, the solution of the HL-LA-2 reformulation provides the upper bound of the congestion cost for hub links, compared to the original model [M1]. □

S.10. Tables S-2—S-5

Table S-2. Average CPU time (s) and optimality gap (%) under the HL-LA-1 and different reformulations for the congestion cost of hubs (AP data set).

$\lvert N\rvert$	c_k^0 & $c_{k,m}^1$	HL-LA-1/H-LA		HL-LA-1/H-SOCP-EL		HL-LA-1/H-SOCP-EP		HL-LA-1/H-SOCP	
		CPU time (s)	Gap (%)	CPU time (s)	Gap (%)	CPU time (s)	Gap (%)	CPU time (s)	Gap (%)
10	LL	0.61	0	0.69	0	**0.19**	0	0.21	0
	LH	0.57	0	0.29	0	**0.16**	0	0.16	0
	HL	0.58	0	0.20	0	**0.20**	0	0.27	0
	HH	0.59	0	0.21	0	0.18	0	**0.16**	0
15	LL	**0.81**	0	1.54	0	1.64	0	1.38	0
	LH	**0.75**	0	1.69	0	1.90	0	1.58	0
	HL	**0.98**	0	1.35	0	1.97	0	2.25	0
	HH	**0.86**	0	2.25	0	1.86	0	1.33	0
20	LL	13.75	0	**12.27**	0	12.47	0	14.03	0
	LH	**8.17**	0	14.04	0	13.75	0	9.62	0
	HL	24.79	0	21.51	0	**18.89**	0	129.44	0
	HH	17.78	0	24.35	0	**17.22**	0	40.19	0
25	LL	**45.16**	0	64.03	0	55.35	0	82.01	0
	LH	**81.50**	0	130.32	0	105.36	0	99.49	0
	HL	**126.30**	0	197.12	0	162.82	0	1773.27	0
	HH	**159.52**	0	171.88	0	233.39	0	2290.11	0
40	LL	**6393.48**	0	10800	8.35	9429.38	5.78	10800	6.89
	LH	**432.53**	0	1606.92	0	679.24	0	8149.63	1.97
	HL	**10800**	9.07	10800	14.34	10800	12.54	10800	27.31
	HH	**1388.93**	0	9290.99	10.42	6321.73	7.97	10800	17.24
50	LL	974.15	0	2757.64	0	**209.98**	0	4961.61	0
	LH	557.71	0	949.23	0	**451.55**	0	1322.28	0
	HL	**7966.92**	1.18	10540.15	3.37	10800	6.67	10800	13.55

	HH	348.57	0	689.27	0	**299.96**	0	10800	4.49
75	LL	10800	--	10800	--	10800	--	10800	--
	LH	10800	--	10800	--	**10800**	65.40	10800	70.42
	HL	10800	--	10800	--	10800	--	10800	--
	HH	10800	--	10800	--	**10800**	67.27	10800	--

Table S-3. Average CPU time (s) and optimality gap (%) under the HL-LA-2 and different reformulations for the congestion cost of hubs (AP data set).

$\|N\|$	c_k^0 & $c_{k,m}^1$	HL-LA-2/H-LA		HL-LA-2/H-SOCP-EL		HL-LA-2/H-SOCP-EP		HL-LA-2/H-SOCP	
		CPU time (s)	Gap (%)	CPU time (s)	Gap (%)	CPU time (s)	Gap (%)	CPU time (s)	Gap (%)
10	LL	**0.23**	0	0.79	0	0.74	0	0.71	0
	LH	**0.17**	0	0.32	0	0.31	0	0.33	0
	HL	**0.19**	0	0.25	0	0.34	0	0.36	0
	HH	**0.18**	0	0.20	0	0.25	0	0.22	0
15	LL	**1.16**	0	4.01	0	5.11	0	5.90	0
	LH	**1.07**	0	2.61	0	3.55	0	3.82	0
	HL	**1.73**	0	6.29	0	9.85	0	15.00	0
	HH	**2.01**	0	9.80	0	23.18	0	199.42	0
20	LL	2758.81	12.60	10800	--	**1087.81**	0	10800	--
	LH	**55.11**	0	2007.98	0	3929.62	0	10800	--
	HL	**37.49**	0	3490.45	0	10800	--	8397.80	0.26
	HH	2743.71	17.38	6419.68	0.34	**3657.73**	0.13	10800	--
25	LL	**8876.74**	34.60	10800	--	10800	--	10800	--
	LH	**203.68**	0	7896.66	0.38	10800	--	10800	--
	HL	**4089.85**	11.64	10800	25.10	10800	--	10800	--
	HH	**8103.25**	48.40	10800	--	10800	--	10800	--
40	LL	**8170.25**	44.90	10800	--	10800	--	10800	--
	LH	10800	--	10800	--	10800	--	10800	--
	HL	**6090.38**	21.02	10800	--	10800	--	10800	--
	HH	**13.80**	0	10800	--	10800	--	10800	--
50	LL	**5822.25**	20.51	10800	--	10800	--	10800	--
	LH	**9.25**	0	10800	--	10800	--	10800	--
	HL	**10800**	57.70	10800	--	10800	--	10800	--
	HH	10800	--	10800	--	10800	--	10800	--
75	LL	**10800**	68.40	10800	--	10800	--	10800	--
	LH	**83.07**	0	10800	--	10800	--	10800	--
	HL	**10800**	56.40	10800	--	10800	--	10800	--
	HH	**71.22**	0	10800	--	10800	--	10800	--

Table S-4. Average CPU time (s) and optimality gap (%) under the HL-SOCP and different reformulations for the congestion cost of hubs (AP data set).

$\lvert N\rvert$	c_k^0 & $c_{k,m}^1$	HL-SOCP/H-LA		HL-SOCP/H-SOCP-EL		HL-SOCP/H-SOCP-EP		HL-SOCP/H-SOCP	
		CPU time (s)	Gap (%)	CPU time (s)	Gap (%)	CPU time (s)	Gap (%)	CPU time (s)	Gap (%)
10	LL	**0.25**	0	0.67	0	0.28	0	0.33	0
	LH	**0.15**	0	0.20	0	0.17	0	0.17	0
	HL	**0.16**	0	0.23	0	0.20	0	0.83	0
	HH	**0.15**	0	0.25	0	0.19	0	0.32	0
15	LL	**1.28**	0	2.25	0	2.00	0	2.42	0
	LH	2.14	0	2.04	0	**2.01**	0	2.27	0
	HL	**2.27**	0	2.46	0	2.32	0	3.62	0
	HH	2.34	0	2.15	0	**2.08**	0	2.81	0
20	LL	24.69	0	28.27	0	**14.08**	0	20.39	0
	LH	14.81	0	20.57	0	17.30	0	**11.59**	0
	HL	59.53	0	27.77	0	**27.56**	0	283.76	0
	HH	32.35	0	23.54	0	**21.75**	0	89.92	0
25	LL	109.83	0	88.20	0	**76.12**	0	86.87	0
	LH	88.91	0	133.55	0	99.14	0	**68.02**	0
	HL	504.23	0	344.97	0	**224.02**	0	2184.49	0
	HH	223.29	0	**187.10**	0	303.81	0	2128.97	0
40	LL	10800	4.70	**4525.22**	0	5330.92	0	10800	7.06
	LH	1334.57	0	1360.44	0	**1275.84**	0	3881.75	0
	HL	10800	10.90	10800	12.20	**10800**	8.72	10800	29.50
	HH	**3897.89**	0	4768.08	0.90	9118.57	0.16	10800	28.10
50	LL	806.88	0	255.43	0	**185.49**	0	10800	1.04
	LH	5534.38	0.11	3790.40	0	**717.01**	0	3130.49	0.29
	HL	10800	4.31	**8467.22**	1.97	10800	5.31	10800	13.30
	HH	807.16	0	**723.41**	0	1919.53	0	9897.61	6.53
75	LL	10800	--	10800	--	10800	--	10800	--
	LH	10800	--	**10800**	61.20	10800	--	10800	--
	HL	10800	--	10800	--	10800	--	10800	--
	HH	10800	--	10800	--	10800	--	10800	--

Table S-5. Experimental results of cost components and congestion levels under different settings of α, c_k^0 and $c_{k,m}^1$ (AP, $\lvert N\rvert = 25$).

c_k^0 & $c_{k,m}^1$	α	Cost components						Congestion levels	
		f ($\times 10^4$)	f_1 ($\times 10^4$)	f_2 ($\times 10^4$)	f_3 ($\times 10^4$)	f_4 ($\times 10^4$)	f_5 ($\times 10^4$)	Hubs	Hub links
LL	0.2	1261.26	424.32	541.12	207.35	41.22	47.26	137.39	4725.70
	0.4	1298.23	424.32	541.12	244.31	41.22	47.26	137.39	4725.70
	0.6	1335.19	424.32	541.12	281.27	41.22	47.26	137.39	4725.70
	0.8	1372.15	424.32	541.12	318.24	41.22	47.26	137.39	4725.70
LH	0.2	2980.14	462.64	467.44	531.33	28.06	1490.70	93.53	3726.69

	0.4	3008.37	462.64	467.44	559.56	28.06	1490.70	93.53	3726.69
	0.6	3036.60	462.64	467.44	587.79	28.06	1490.70	93.53	3726.69
	0.8	3064.84	462.64	467.44	616.02	28.06	1490.70	93.53	3726.69
HL	0.2	1637.26	587.52	512.56	281.77	206.12	49.29	13.74	4929.70
	0.4	1676.53	587.52	512.56	321.04	206.12	49.29	13.74	4929.70
	0.6	1715.80	587.52	512.56	360.31	206.12	49.29	13.74	4929.70
	0.8	1755.07	587.52	512.56	399.58	206.12	49.29	13.74	4929.70
HH	0.2	3481.04	509.56	456	536.63	344	1634.81	22.93	4087
	0.4	3512.03	509.56	456	567.62	344	1634.81	22.93	4087
	0.6	3543.03	509.56	456	598.62	344	1634.81	22.93	4087
	0.8	3574.02	509.56	456	629.61	344	1634.81	22.93	4087